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Write a program to implement Logistic Regression (LR) algorithm in python

CODE:

```
import numpy as np

X = np.array([
    [2, 60],
    [3, 65],
    [4, 70],
    [5, 75],
    [6, 80],
    [7, 85],
    [8, 90],
    [9, 95]
], dtype=float)

# Target: 0 = Fail, 1 = Pass
y = np.array([0, 0, 0, 1, 1, 1, 1, 1], dtype=float)

X[:, 0] = X[:, 0] / 10
X[:, 1] = X[:, 1] / 100

X = np.c_[np.ones(X.shape[0]), X]

weights = np.zeros(X.shape[1])

def sigmoid(z):
    return 1 / (1 + np.exp(-z))

learning_rate = 0.1
epochs = 1000

for i in range(epochs):
    z = np.dot(X, weights)
    predictions = sigmoid(z)
    gradient = np.dot(X.T, (predictions - y)) / len(y)
    weights = weights - learning_rate * gradient
probabilities = sigmoid(np.dot(X, weights))
```

```
predictions = (probabilities >= 0.5).astype(int)
accuracy = np.mean(predictions == y) * 100
print("Final Weights:")
print(weights)
print("\nPredicted Probabilities:")
print(probabilities)
print("\nPredicted Classes:")
print(predictions)
print("\nActual Classes:")
print(y.astype(int))
print("\nAccuracy: {:.2f}%".format(accuracy))
```

OUTPUT

```
Final Weights:
[-2.58661686  4.66779248  1.04058781]

Predicted Probabilities:
[0.26333511 0.37521766 0.50222789 0.62895124 0.74010779 0.
 82711792
 0.88935269 0.9310515 ]

Predicted Classes:
[0 0 1 1 1 1 1 1]

Actual Classes:
[0 0 0 1 1 1 1 1]

Accuracy: 87.50%

=== Code Execution Successful ===
```